

ARIN5102 2025–26 Fall Semester Project

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Problem 1. (20%) *Due by Thursday, November 20 on canvas.*

Context. This project is on your use and understanding of GPT-4o. While not one of the best LLMs out there anymore, it is still one the better ones that are available on `genai.ust.hk`, which can also be accessed through API on azure.

Task. Provide 5 examples of problematic behaviours by GPT-4o. These could be incorrect or inappropriate answers to your questions, or correct answers but incorrect reasoning behind the answers. You should be able to find some examples on the internet. You are allowed to re-use them. However, if you do, you must refer to the source. Many of them are math-related, like counting the number of characters in a sentence. Try to be innovative. For example, can you come up with an example that shows GPT-4o *simplify* did not understand your questions?

Evidence. Each of your examples must be accompanied by the answers from GPT-4o as provided by our `genai.ust.hk`, either through browser or through API, *and* your analyses of why the answers are problematic.

How to Use This Template

For each example below, fill in:

1. **Question** you asked GPT-4o (plain text).
2. **Model Answer** you received (plain text).
3. **API Raw JSON** (copy the full request/response JSON if available).
4. **Screenshot/Image** (paste the full request/response figure if available).
5. **Analysis** explaining what is wrong and why.

1 Example 1: Counting the number of letters

Question (Plain Text)

-Write a sentence that contains the letter f exactly three times

Model Answer (Plain Text)

-The fluffy cat swiftly leapt off the fence.
(Wrong answer about the requirement)

API Raw JSON

Actually, it is extremely hard to extract the conversation part from the Raw JSON file, so I just show Example 1 here and upload all the JSON files with my report. But we have the screenshots as the evidence. Appreciate your understanding.

Listing 1: GPT-4o API raw JSON (request/response)

```
1 [{"id": "8ecabec6-1a5c-4649-a9f2-87a9bae98cb9",
2  "user_id": "9629e530-d25b-4050-984d-dc5034fd7c72",
3  "title": "Sentence with 3 Fs",
4    "chat": {
5      "id": "", "title": "Sentence with 3 Fs",
6      "models": ["gpt-4o"],
7      "params": {},
8      "history": {
9        "messages": {
10         "75d3223c-1d6b-4793-a6b0-b8278cf49307": {
11           "id": "75d3223c-1d6b-4793-a6b0-b8278cf49307", "parentId": null,
12           "childrenIds": ["b419ed5c-5d52-44b2-8276-710061cb3f96"],
13           "role": "user",
14           "content": "Write a sentence that contains the letter f exactly
15             three times",
16           "timestamp": 1763287656,
17           "models": ["gpt-4o"]
18         },
19         "b419ed5c-5d52-44b2-8276-710061cb3f96": {
20           "parentId": "75d3223c-1d6b-4793-a6b0-b8278cf49307", "id": "b419ed5c-5d52-44b2-8276-710061
21             cb3f96", "childrenIds": [],
22           "role": "assistant",
23           "content": "The fluffy cat swiftly leapt off the fence.",
24           "model": "gpt-4o",
25           "modelName": "gpt-4o | Multi-Modal", "modelIdx": 0, "timestamp": 176328
26             7656,
27           "done": true
28         }
29       }
30     }
31   }
```

Analysis

This is a broader topic. The fundamental reason is that LLMs like GPT do not directly recognize individual letters when processing text; instead, they break the text down into tokens. This tokenization strategy is significantly different from human word recognition. Because it disrupts the

complete sequence of letters, there exists a natural obstacle to artificial intelligence when counting letters.

Screenshot / Image Evidence

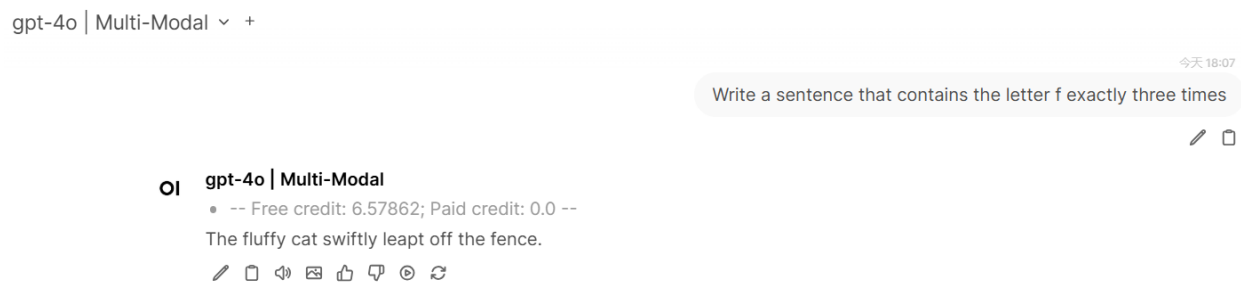


Figure 1: Evidence screenshot for Example 1.

2 Example 2: Character discrimination

Question and Model Answer(Plain Text)

-Q: Write a sentence that starts and ends exactly with the same letter f

-A: Frogs leap freely in the forest.

-Q: Write a sentence that starts exactly with the letter f

-A: Fresh flowers filled the room with a delightful fragrance.

-Q: Write a sentence that ends exactly with the letter f

-A: She wore a cozy scarf made of soft wool.

(Failed in Q1 & Q3)

Screenshot / Image Evidence

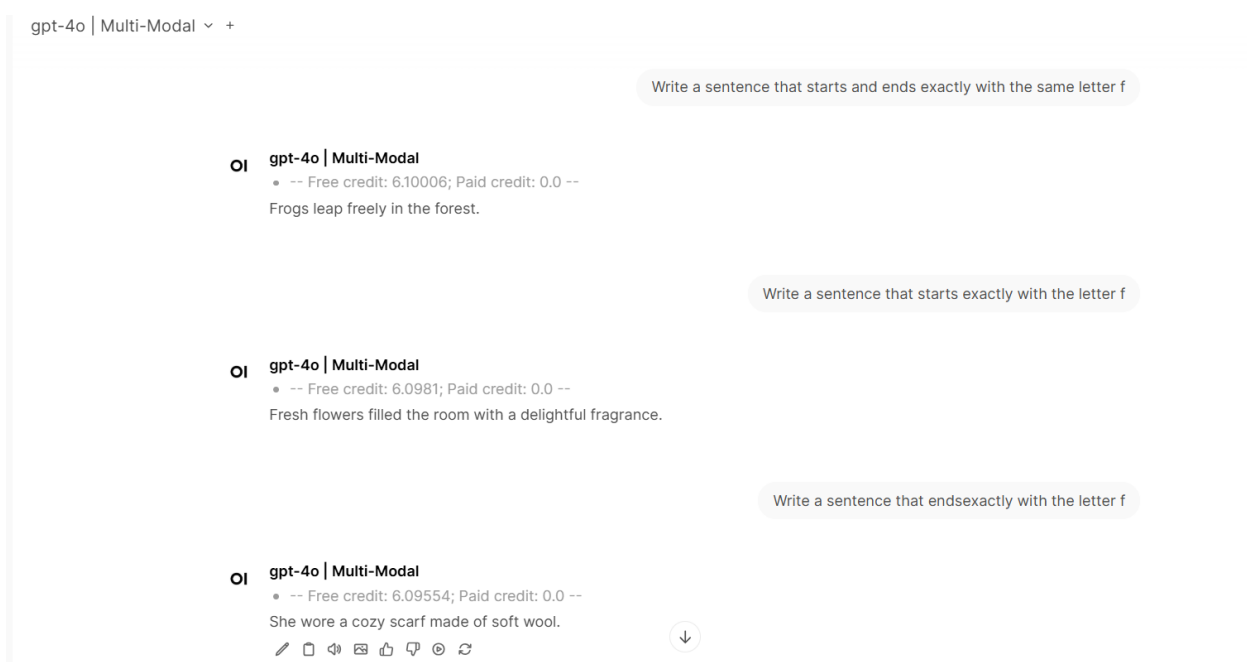


Figure 2: Evidence screenshot for Example 2.

Analysis

Extension of Example 1, we can observe that GPT-4o doesn't struggle with sentence beginnings. But after it starts with a particular word, continuation writing becomes affected by both the sentence's meaning itself and the limitations imposed by the ending, often leading to failure.

3 Example 3: Database update

Question (Plain Text)

-What's the date today?

Model Answer (Plain Text)

-Today's date is October 21, 2023.
(Wrong date when the conversation happens)

Screenshot / Image Evidence



Figure 3: Evidence screenshot for Example 3.

Analysis

This is clearly related to the built-in database of GPT-4o, where dates are locked to a specific time when the latest update was finished, and cannot go on with time of reality. In fact, this leads to a series of incorrect question-and-answer statements about the real world. I will only show one representative example at Example 4.

4 Example 4: Outdated reality

Question and Model Answer(Plain Text)

-Q: Please tell me which season of the NBA league is in progress?

-A: As of October 21, 2023, the 2023-2024 NBA season is in progress. The regular season typically begins in October and runs through April, followed by the playoffs, which conclude with the NBA Finals in June.(Expected '2025-2026')

-Q: And which team is the defending champion at today?

-A: As of October 21, 2023, the Denver Nuggets are the defending NBA champions. They won the 2023 NBA Finals, securing their first championship in franchise history.(The truth is Thunder)

Screenshot / Image Evidence

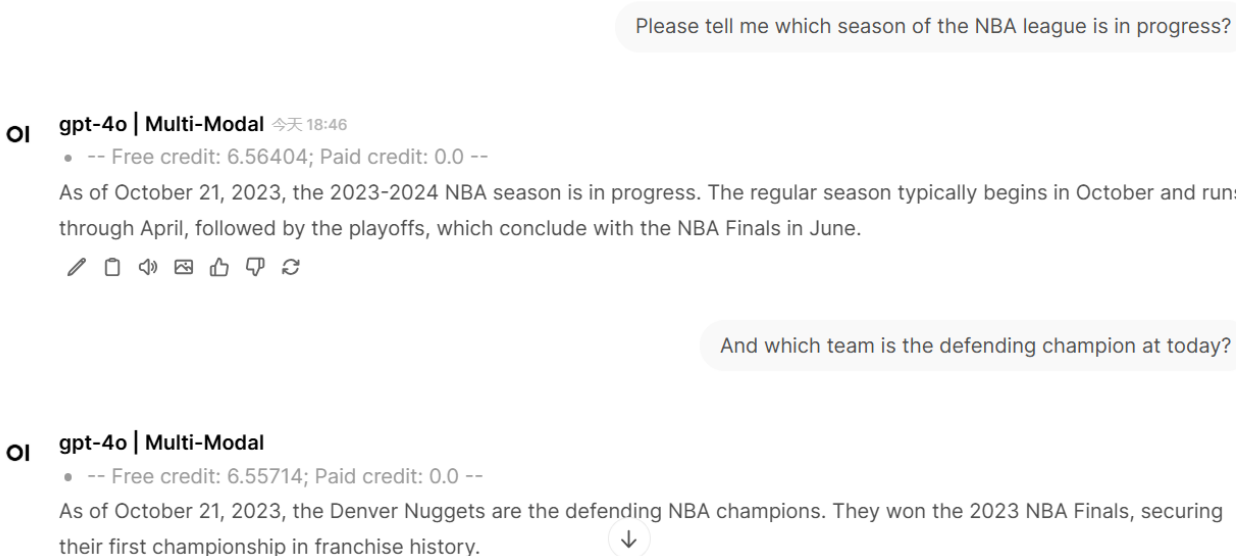


Figure 4: Evidence screenshot for Example 4.

Analysis

As we mentioned earlier, outdated data and disability to searching for new ones prevent GPT-4o from getting connection with the real-world. Most emerging LLMs on the market can handle this task, but actually asking them for all the news from different fields is also quite difficult.

5 Example 5: Fictitious reality

Question and Model Answer(Plain Text)

-Q: how many sons had eise eisinga in total

-A: Eise Eisinga, the Dutch amateur astronomer known for building the Eise Eisinga Planetarium, had three sons.

-Q: how many kids had eise eisinga in total

-A: Eise Eisinga had a total of four children.

-Q: what are the sexual of them

-A: Eise Eisinga had three sons and one daughter.

This question is from a Twitter sharing. Here is the truth about Eise Eisinga from Wikipedia: *At the age of 24, he married Pietje Jacobs[3] (1748[4] – 18 July 1788)[5] and they had **three children, one girl and two boys**. Meanwhile, his wife died, and on 27 May 1792, he married Trijntje Elkes Sikkema (21 February 1764) in Visvliet. They had **one son and two daughters**.*

Screenshot / Image Evidence

gpt-4o | Multi-Modal ▾ +

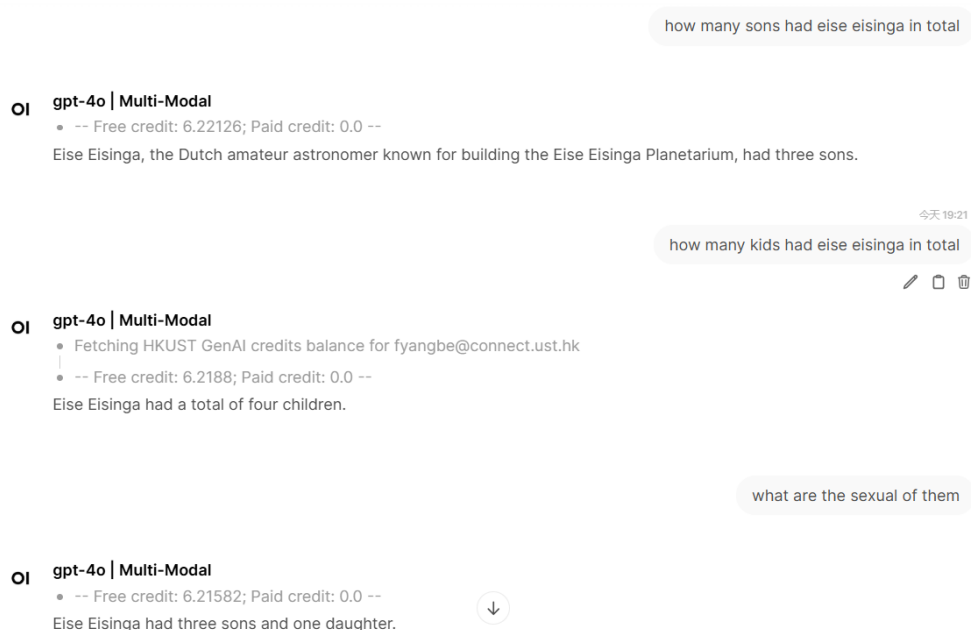


Figure 5: Evidence screenshot for Example 5.

Analysis

Unlike Example 4, fictional rewriting of history may come from information chaos in the database or an uncorrected search pattern. In this case, the GPT-4o may initially retrieve all the correct information (3 sons + 3 daughters), but was only asked to answer a narrow range of information. However, when asked to broaden the scope, it searched again, only found a daughter from the first marriage and overwrote the former results about daughters. Then GPT-4o added 1 to the first answer, yielding the wrong answer.

6 Example 6: Inconsistency and defect in algorithms

Question and Model Answer(Plain Text)

This example is actually from an attempt programming of another course. Since the conversation is much longer, I will show it in the following image Figure 6.

Screenshot / Image Evidence

Analysis

As I pointed in the conversation, when given 'google', the generative algorithm output 'googog' which is undoubtedly not a Palindrome string. Looking back to the algorithm, it seems that GPT-4o indeed understand the meaning of 'Palindrome' and how to convert 'google' into one like that: divide into two parts-try to match the two parts-combine them to a new caption. However, the error always happened in the last step. There were no failure when the GPT-4o handled with matching task of two single letters, but it met trouble in having memory of the original caption for the rewritten work.

do you know about the lexicographically smallest caption

OI gpt-4o | Multi-Modal

• -- Free credit: 5.37836; Paid credit: 0.0 --

Yes, the lexicographically smallest caption refers to the arrangement of words or characters in a sequence that would appear first if sorted alphabetically. In the context of captions, this means selecting words or phrases that, when ordered alphabetically, would come before any other possible combination of words or phrases. This concept is often used in programming and algorithm design to ensure that a sequence is organized in the most efficient or standardized way. If you have a specific context or example in mind, feel free to share more details!

Design an algorithm to detect whether a given string is a palindrome. If it is not a palindrome, rewrite it as a string, which is a palindrome and of equal length, with the minimum in lexicographical cost. Then you verify your algorithm with 'google'

OI gpt-4o | Multi-Modal

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To design an algorithm that detects whether a given string is a palindrome and, if not, rewrites it as a palindrome with the minimum lexicographical cost, follow these steps:

Algorithm

- Check if the String is a Palindrome:**
 - Compare the string with its reverse. If they are the same, the string is already a palindrome.
- Convert to Lexicographically Smallest Palindrome:**
 - If the string is not a palindrome, create a new string that is a palindrome and has the minimum lexicographical cost.
 - Initialize two pointers: one at the start (`left`) and one at the end (`right`) of the string.
 - Iterate while `left` is less than `right` :
 - Compare the characters at `left` and `right` .
 - If they are not the same, replace the character at the `right` with the character at `left` if `left` is lexicographically smaller, otherwise replace the character at `left` with the character at `right` .
 - Move the `left` pointer one step to the right and the `right` pointer one step to the left.
 - Continue this process until the pointers meet or cross.
- Output the Result:**
 - If the original string was a palindrome, output it.
 - Otherwise, output the newly formed palindrome.

Verification with 'google'

Let's apply the algorithm to the string "google":

- Check if "google" is a Palindrome:**
 - "google" is not equal to its reverse "elgoog", so it is not a palindrome.
- Convert "google" to Lexicographically Smallest Palindrome:**
 - Initialize `left` at index 0 and `right` at index 5.
 - Compare 'g' (left) and 'e' (right). Replace 'e' with 'g' (since 'g' < 'e').
 - Compare 'o' (left) and 'l' (right). Replace 'l' with 'o' (since 'o' < 'l').
 - Compare 'o' (left) and 'g' (right). Replace 'g' with 'o' (since 'o' < 'g').
 - The resulting palindrome is "googog".
- Output:**
 - The lexicographically smallest palindrome of equal length to "google" is "googog".

This algorithm efficiently checks and transforms a string into the desired palindrome form.

Figure 6: Evidence screenshot for Example 6.

References

- [1] Some inspirations motivated by <https://github.com/giuven95/chatgpt-failures?tab=readme-ov-file> (accessed: November 16, 2025).